Neuromorphic Computing

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Abstract

Modern deep learning has produced transformative capabilities, but current large models and conventional computing architectures face important limitations: high energy use, high latency in on-device contexts, and poor alignment with how biological brains compute. Neuromorphic computing and spiking neural networks (SNNs) are brain-inspired approaches that promise event-driven, low-power, and temporally aware computation. This literature review synthesizes technical research across three themes: (1) shortcomings of the von Neumann model when applied to AI workloads, (2) adaptability and training challenges specific to SNNs (including surrogate gradient methods, STDP, and ANN→SNN conversion), and (3) practical obstacles for neuromorphic hardware adoption (programming ecosystems, mapping algorithms to silicon, device non-idealities, and economics). The review integrates results from recent surveys and research papers to highlight where progress has been made and where gaps remain. The synthesis indicates that algorithm-hardware co-design, improved training methods, robust benchmarking, and open toolchains are necessary to realize neuromorphic systems at scale.

Keywords: neuromorphic computing, spiking neural networks, von Neumann, surrogate gradients, hardware–software co-design.

Introduction

Artificial intelligence has made rapid progress over the last decade, driven primarily by large artificial neural network (ANN) models running on traditional von Neumann hardware (CPUs/GPUs). However according to Kimovski et al. (2024), while this approach has produced strong performance on many benchmarks, the prevailing compute model exhibits several structural weaknesses for the next generation of AI applications: data movement between memory and processor causes a power and performance bottleneck; synchronous, dense compute is inefficient for sparse, event-driven workloads; and present hardware lacks native support for temporal/spike-based computation inspired by biology. These constraints motivate the investigation of neuromorphic computing: hardware and algorithms that mimic neuronal and synaptic dynamics, executing spike-based networks with high parallelism and low power consumption (Al Abdul Wahid et al., 2024).

This literature review focuses on the technical intersection of Spiking Neural Networks (SNNs) and Neuromorphic Computing (NC). It is organized around three themes. Theme 1 reviews the principal limitations of traditional von Neumann architectures for AI workloads. Theme 2 surveys the adaptability and training challenges of SNNs, including methods developed to close the gap between SNNs and ANNs. Theme 3 addresses hardware and system challenges associated with neuromorphic computing platforms, including mapping algorithms to silicon and production-level constraints. Taken together, the evidence suggests that algorithm–hardware co-design, open and reproducible toolchains, and energy-aware benchmarks are the most actionable priorities. Advancing these areas will make neuromorphic systems more competitive for production deployments in autonomous systems, robotics, and always-on edge sensing.

Limitations of Traditional Computing

Modern computing systems are dominated by a Von Neumann architecture that separates memory and computation. Muralidhar et al. (2022) describes the von Neumann bottleneck as - the cost (time and energy) of shuttling data between memory and processor, this becomes a primary limiter when workloads are memory-bound or when models require vast parameter accesses. In case of AI, especially large models and streaming sensory processing, the cost of memory movement can dwarf arithmetic costs, increasing latency and energy expenditure.

Another limitation is the mismatch between synchronous dense computation on GPUs and the sparse, asynchronous nature of many real-world signals. For applications that produce sparse events (e.g. changes in a scene captured by event-based cameras), traditional frame-based processing wastes computation on unchanged regions and cannot take full advantage of temporal sparsity. In contrast, spike-based processing is inherently event-driven and can avoid unnecessary operations.

Thermal and energy constraints also shape system design. Kimovski et al. (2024) notes high-performance GPUs expend significant energy and produce heat, challenging continuous on-device operation for robotics, battery-powered sensors, and edge devices. Muralidhar et al. (2022) add that such physical constraints motivate computing models that merge memory and computation and leverage sparse event processing to reduce energy per useful operation.

Thus, according to Davies et al. (2021) biological plausibility is a practical consideration, because biological nervous systems perform robust, online, low-power processing using local plasticity rules and massively parallel sparse communication. Translating these principles into silicon offers potential efficiency and latency benefits, but doing so requires departing from von Neumann design assumptions. The literature consistently argues that certain future AI domains

such as real-time robotics, always-on sensing, and adaptive control will be better supported by neuromorphic paradigms than by traditional architectures (Chowdhury et al., 2025).

Synthesis of Limitations of Traditional Computing highlights that -current AI is limited by the foundational structure of von Neumann computing, overcoming them calls for architectures that minimize memory-movement, embrace asynchronous event processing, and co-locate memory and computation which precisely the goals of neuromorphic approaches, which we'll discuss in the next theme.

Adaptability and Training Challenges of Spiking Neural Networks (SNNs)

SNNs model information using discrete spikes and time; neurons fire only when membrane potentials cross thresholds, and timing encodes information. This temporal, event-driven representation is a key advantage for low-latency and energy-efficient processing. Nevertheless, SNNs face significant training and adaptability challenges that have prevented them from easily attaining the performance of modern ANNs on large benchmarks.

I. Training difficulty and non-differentiability

Standard gradient-based learning relies on differentiable activation functions. Spikes are discontinuous events, and thus direct backpropagation through spiking dynamics is not straightforward. However there are several approaches have been proposed.

Surrogate gradient methods by Gouda et al. (2024) and Li et al. (2024), where we replace the non-differentiable spike function with a smooth surrogate derivative during the backward pass, allowing gradient descent training of SNNs end-to-end. Surrogate gradients have enabled deeper spiking architectures to be trained and have improved task performance but tuning and stability remain areas of active research. On contrast, Gao et al. (2023) and Wang et al. (2024)

proposed ANN-to-SNN conversion -train an ANN with mature techniques, then convert activations/weights to spike rates in an SNN. This approach often preserves accuracy but can require long simulation windows to approximate real-valued activations, which reduces latency advantages.

II. Scalability and performance gaps

Despite progress, SNNs often trail ANNs on large, complex datasets when trained from scratch. This gap is narrowing with surrogate gradients and hybrid schemes, but Zhou et al. (2024) from Frontiers review notes that achieving parity without sacrificing latency and energy benefits is still a research target. The literature suggests that new spiking architectures, not mere one-to-one translations of ANN topologies, are needed to exploit temporal coding fully.

III. Tooling and frameworks

Zhou et al. (2024) highlights one of the main obstacles in SNN adoption -the fragmented SNN ecosystem. SNNs do have simulation frameworks such as Brian2, Nengo, BindsNET, and SNN libraries but they are built atop PyTorch, but there is no single dominant, user-friendly ecosystem comparable to TensorFlow or PyTorch built for ANNs. This fragmentation complicates reproducibility and slows adoption by mainstream ML practitioners.

This synthesis highlights that training and adaptability challenges for SNNs are both algorithmic (non-differentiability, architecture design) and infrastructural (tooling, benchmarks). Progress in surrogate gradients, hybrid training, and community toolchains is promising, but converging to robust, scalable methods that preserve SNNs' energy/latency advantages remains a primary research priority. The next theme discusses how Neuromorphic hardware compliments SNNs in solving the limitations of Von Neumann computing.

Challenges with Neuromorphic Computing

Neuromorphic computing seeks to realize SNN advantages in hardware. Several commercially developed and research platforms exist (Intel Loihi family, IBM TrueNorth, SpiNNaker, Tianjic, BrainScaleS), each reflecting different trade-offs between programmability, energy, and scalability (Davies et al., 2021; Zhang et al., 2023). Nonetheless, bringing neuromorphic hardware into widespread use faces multiple obstacles.

I. Hardware heterogeneity and device non-idealities

Neuromorphic platforms vary: some are digital (Loihi), some analog or mixed-signal, and emerging devices (memristors, spintronics, photonic devices) promise higher density and lower power but introduce device variability and analog mismatch issues (Muir et al., 2025). Device non-idealities (noise, drift, limited precision) complicate mapping algorithms and require resilience at the algorithm level.

II. Software and programming model fragmentation

Davies et al, (2021) add that there is no unified programming model for neuromorphic systems. Porting an SNN from a simulator to a specific chip often requires substantial modification and hardware-aware mapping. This raises a steep learning curve for developers and reduces reproducibility. Middleware and abstraction layers that present consistent APIs to researchers are beginning to emerge but are not yet mature.

III. Economic and manufacturing issues

Most neuromorphic devices are research prototypes or low-volume products. Scaling manufacturing to commercial volumes, reducing cost, and building a supporting ecosystem (tools, libraries, trained engineers) are nontrivial challenges. The commercialization path requires demonstrating clear application value (e.g., energy savings at scale, unique capabilities) that justify investment (Muir et al., 2025).

IV. Benchmarking and evaluation

Al Abdul Wahid et al. (2024) add, the field lacks standardized, energy-aware benchmarks for SNN/neuromorphic systems. Benchmarks that combine accuracy, latency, and energy per inference - and that are compatible with event-based tasks - are essential to compare approaches fairly and to drive optimization.

Synthesis of Neuromorphic hardware offers strong theoretical benefits, but to move from lab prototypes to production systems the community must address heterogeneity and device reliability, create unified programming abstractions, optimize mapping and partitioning, and establish standardized benchmarks and metrics that reflect operational constraints (energy, latency, robustness).

Conclusion

Neuromorphic computing and spiking neural networks present a promising route toward energy-efficient, temporally sensitive AI suited for real-time and edge applications. The literature shows clear conceptual advantages: event-driven processing, the ability to exploit temporal coding, and potential orders-of-magnitude reductions in energy per operation. However, the path to broad practical adoption traverses significant obstacles in three areas: first, the structural limitations of von Neumann hardware for AI workloads motivate alternative architectures; second, SNNs still contend with training and scalability problems that require improved surrogate gradient techniques, hybrid training, and novel architectures; third, neuromorphic hardware must become more accessible and reliable through standardized software stacks, robust mapping methods, device engineering, and benchmarking.

Taken together, the evidence suggests that algorithm–hardware co-design, open and reproducible toolchains, and energy-aware benchmarks are the most actionable priorities. Advancing these areas will make neuromorphic systems more competitive for production deployments in autonomous systems, robotics, and always-on edge sensing. The interdisciplinary nature of this research - spanning device physics, VLSI, machine learning, and computational neuroscience - requires collaborative teams and coordinated community standards to realize neuromorphic computing's promise.

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